

Prompt: Write a C program that will pick a secret number from 1 to 10 within 3 guesses and prints a win or lose stopping immediatly if the user guesses correctly that is fixed, uses printf and scanf to get inputs, a for loop, if-else statements to check if guess is too high low or correct

Claude:

```
c
#include <stdio.h>

int main() {
    int secret = 7; // fixed secret number
    int guess;
    int won = 0;

    printf("Guess the secret number (1-10). You have 3 tries.\n");

    for (int attempt = 1; attempt <= 3; attempt++) {
        printf("Attempt %d: Enter your guess: ", attempt);
        scanf("%d", &guess);

        if (guess == secret) {
            won = 1;
            break; // stop immediately on correct guess
        } else if (guess > secret) {
            printf("Too high!\n");
        } else {
            printf("Too low!\n");
        }
    }

    if (won) {
        printf("You win! The secret number was %d.\n", secret);
    } else {
        printf("You lose! The secret number was %d.\n", secret);
    }

    return 0;
}
```

Copilot:

```
1 #include <stdio.h>
2
3 int main() {
4     int secret = 7;    // Fixed secret number
5     int guess;
6     int i;
7
8     printf("Guess the secret number (1-10).\n");
9     printf("You have 3 guesses.\n");
10
11     for (i = 1; i <= 3; i++) {
12         printf("Guess %d: ", i);
13         scanf("%d", &guess);
14
15         if (guess == secret) {
16             printf("You win! You guessed the secret number.\n");
17             return 0;    // Stop immediately
18         }
19         else if (guess < secret) {
20             printf("Too low.\n");
21         }
22         else {
23             printf("Too high.\n");
24         }
25     }
26
27     printf("You lose! The secret number was %d.\n", secret);
28
29     return 0;
30 }
```

Show less

Analasys: I picked the Copilot code to use due to clearer readability as well as the use of returns instead of breakage that would save on memory for the loop. I accessed Claude through the web going to its website, for Copilot I accessed it through the desktop app. I cleared up some of the text statements and added comments to the code. Comparing the two generations, they are very similar to each other one of the major differences I see is the way they detect winning and display that to the user. I think that that way copilot did it was more effective since it made less variables and returned out instead of breaking the loop.